



MOTION ACTIVATED LIGHTNING SYSTEM

GROUP MEMBERS:

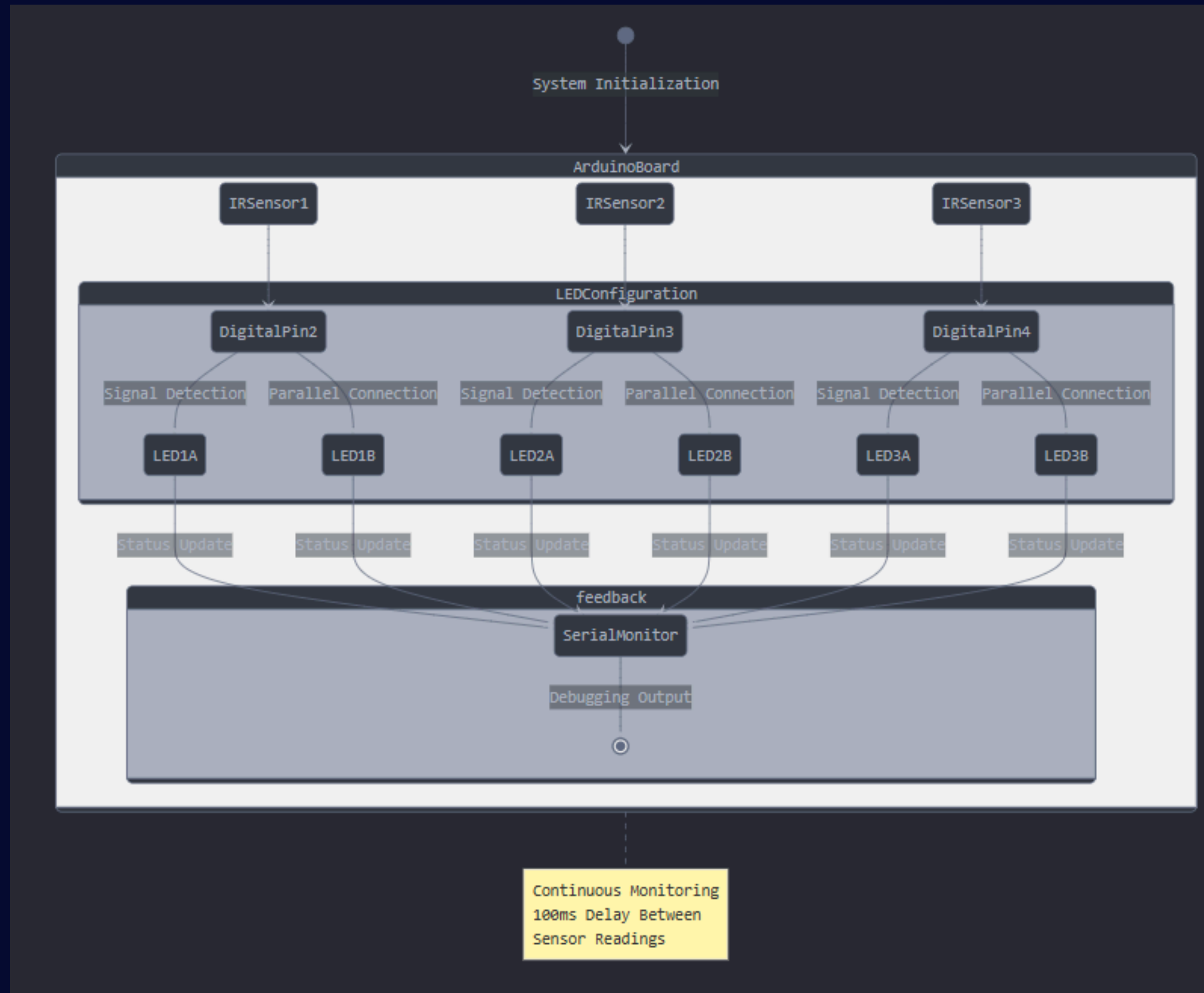
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block diagram



PROBLEM STATEMENT:

In many residential and commercial spaces, lighting systems often remain on unnecessarily when there is sufficient ambient light or when no one is present. This leads to significant energy wastage and increased electricity bills.

PROBLEM SOLUTION:

Design and implement an automated motion-activated lighting system using IR sensors that intelligently adjusts light levels based on motion detection. The system should automatically turn on the lights when motion is detected.

COMPONENTS

ARDUINO
IR SENSORS
LEDS
RESISTORS
JUMPERWIRES



METHODOLOGY ABOUT THE SOLUTION:

- Use IR sensors to detect motion near streetlights.
- Connect IR sensors to a microcontroller (e.g., Arduino) to process signals and control the lights.
- Lights will be either fully ON or OFF, without any brightness adjustment mechanism.

CODE:

```
int irSensorPin1 = 2;
int irSensorPin2 = 3;
int ledPin1 = 7;
int ledPin2 = 6;
void setup() {
  pinMode(irSensorPin1, INPUT);
  pinMode(irSensorPin2, INPUT);
  pinMode(ledPin1, OUTPUT);
  pinMode(ledPin2, OUTPUT);
  Serial.begin(9600);
}
```

```
void loop(){
int sensorState1 = digitalRead(irSensorPin1);
int sensorState2 = digitalRead(irSensorPin2);

    if (sensorState1 == LOW) {
        digitalWrite(ledPin1, HIGH);
        Serial.println("IR signal detected!");
    }
    else {
        digitalWrite(ledPin1, LOW);
        Serial.println("No IR signal detected.");
    }

    if (sensorState2 == LOW) {
        digitalWrite(ledPin2, HIGH);
        Serial.println("IR signal detected!");
    }
    else {
        digitalWrite(ledPin2, LOW);
        Serial.println("No IR signal detected.");
    }

    delay(100);
}
```

RESULT OBTAINED SO FAR:

The streetlights are designed to illuminate whenever motion is detected from a vehicle passing in front of them.

CHALLENGES FACED SO FAR:

- 1) Certain IR sensors have an extended detection range, which may cause them to inadvertently detect motion from vehicles or objects near opposite streetlights.
- 2) It is not feasible to power all the IR sensors solely from a single Arduino board, as the onboard power supply may not support the required current for multiple sensors. An external power source should be considered to ensure stable and efficient operation.